

Making Data Ethics Actionable



March 24, 2026

DRIVENDATA



Jay Qi

Principal Data Scientist
Director of Engineering

Hi, I'm Jay



- Principal Data Scientist and Director of Engineering at DrivenData
- 10+ years in data science, machine learning, and scientific computing
- Active open source maintainer and contributor
 - Data science tools
 - Developer utilities
 - Civic tech applications



Data Science + Social Impact

Data & AI Services • Machine Learning Competitions • Open Source Projects



www.drivendata.org



[@drivendata.org](https://twitter.com/drivendata.org)



github.com/drivendataorg

Agenda

- Why does data ethics matter?
- Checklists + Examples
- **deon**: an open source data ethics toolkit

**Why does data ethics
matter?**

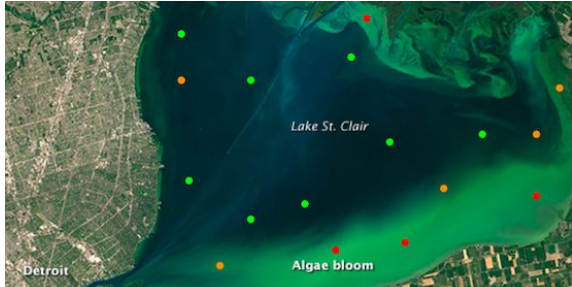
Data is increasingly used to drive decision-making.

Statistical models

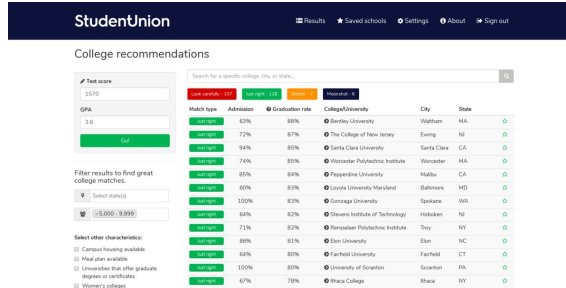
Machine learning

LLMS & Generative AI

We all want to use data for positive outcomes that align with our values...



Monitoring water quality from satellite imagery



The screenshot shows the StudentUnion website interface. It includes a search bar for 'College recommendations' and a table of results. The table has columns for 'Rank type', 'Admission', 'Graduation rate', 'College/university', 'City', and 'State'. The results are filtered by 'Test score' (1570) and 'GPA' (3.8).

Rank type	Admission	Graduation rate	College/university	City	State
100%	82%	88%	Bentley University	Waltham	MA
100%	72%	87%	The College of New Jersey	Ensign	NJ
100%	94%	85%	Santa Clara University	Santa Clara	CA
100%	74%	85%	Worcester Polytechnic Institute	Worcester	MA
100%	85%	84%	Regeneron University	Malibu	CA
100%	80%	83%	James University Maryland	Baltimore	MD
100%	100%	83%	Savanna University	Spokane	WA
100%	64%	82%	University of Technology	Indianapolis	IN
100%	71%	82%	Regeneron Polytechnic Institute	Toy	NY
100%	86%	81%	Elon University	Elon	NC
100%	84%	80%	Fairfield University	Fairfield	CT
100%	100%	80%	University of Southern	Savannah	GA
100%	87%	78%	Utica College	Utica	NY

Matching students with schools where they are likely to succeed



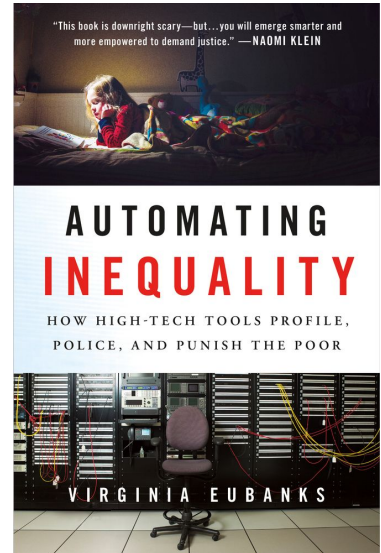
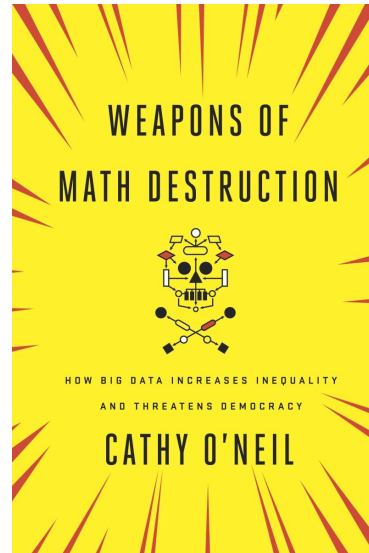
Protecting endangered beluga whales with computer vision

... but those outcomes depend on what is encoded in the data. Intentionality is critical.

Machine Bias

There's software used across the country to predict future criminals. And it's biased against blacks.

by Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, ProPublica
May 23, 2016



**It's not just data scientists who
need to pay attention anymore...**

Data Scientists

**... More people than ever are creating
and operationalizing these systems.**

Software Engineers

Lawyers

Designers

Data Scientists

Policymakers

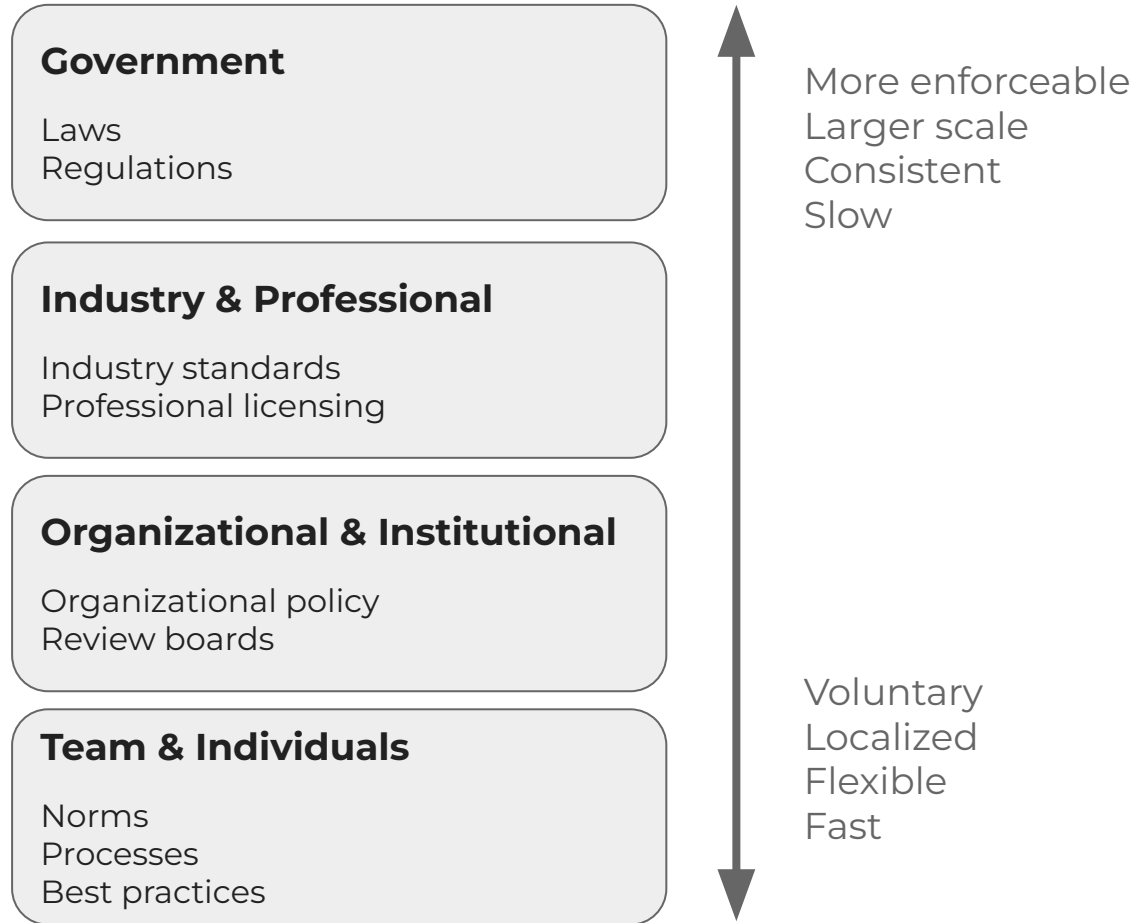
Researchers

Scientists

Teachers

Doctors

**Effective ethics
needs governance
across all layers.**



Effective ethics needs governance across all layers.

Today's talk will focus
on the layer most
actionable to you.



Government

Laws
Regulations

Industry & Professional

Industry standards
Professional licensing

Organizational & Institutional

Organizational policy
Review boards

Team & Individuals

Norms
Processes
Best practices



More enforceable
Larger scale
Consistent
Slow

Voluntary
Localized
Flexible
Fast

An actionable toolkit:

Checklists + Examples

Why an ethics checklist with examples?

Checklists...

- ✓ Connect principles to practice.
- ✓ Are designed to be actionable and embedded in workflows: specific, focused on execution, used repeatedly.

Examples...

- ✓ Demonstrate what it looks like for things to go well or poorly in the real world

Keep in mind...

Ethics is hard.

There is no free lunch.

Tradeoffs are inevitable.

No one right answer.

Reasonable people can disagree.

Good intentions aren't enough.

Must actively consider and anticipate consequences.



Deon is an open source toolkit that allows you to easily add an ethics checklist to your data projects.

<https://deon.drivendata.org>

An open source Python tool

- Easily integrated into a software workflows, familiar to people building with data and AI
- Scriptable
- Customizable
- Support for many formats: .md, .html, .ipynb, .rst, .txt



Use Deon's command-line tool to easily add a checklist to your project as documentation.

```
uvx deon -o ETHICS.md
```

```
uvx deon --help
```

An open source website

- View and download/copy the rendered default checklist.
- Reference for real-world examples connected to each default checklist item.
- Contribute updates or new examples.

About — Deon

deon.drivendata.org/#what-is-deon-designed-to-do

Incognito

Examples

Rendered versions

- [Markdown](#)
- [HTML](#)
- [Jupyter Notebook](#)
- [RST](#)
- [Text](#)

 Check out deon on GitHub



What is deon designed to do?

We created **deon** to help data scientists across the sector be more intentional in their choices, and more aware of the ethical implications of their work. We use that perspective to make decisions about contributions, issues, PRs, and other maintenance and support activities.

1. 🛡️ **Our goal is not to be arbitrators of what ethical concerns merit inclusion.** We have a [process for changing the default checklist](#), but we believe that many domain-specific concerns are not included and teams will benefit from developing [custom checklists](#). Not every checklist item will be relevant. We encourage teams to remove items, sections, or mark items as N/A as the concerns of their projects dictate.
2. 🌍 This checklist is designed to provoke conversations around **issues where data scientists have particular responsibility and perspective**. It's not up to data scientists alone to decide what the ethical course of action is. This has always been a responsibility of organizations that are part of civil society. Conversations should be part of a larger organizational commitment to doing what is right.
3. 💬 Items on the checklist are **meant to provoke discussion** among good-faith actors who take their ethical responsibilities seriously. We are working at a level of abstraction that cannot concretely recommend a specific action (e.g., "remove variable X from your model"). Because of this, most of the items are framed as prompts to discuss or consider. Teams will want to document these discussions and decisions for posterity.
4. 🌐 We believe in the **power of examples** to bring the principles of data ethics to bear on human experience. This repository includes a [list of real-world examples](#) connected with each item in the default checklist. We encourage you to contribute relevant use cases that you believe can benefit the community by their example. In addition, if you have a topic, idea, or comment that doesn't seem right for the documentation, please add it to the [wiki page for this project](#)!

Examples — Deon

deon.drivendata.org/examples/

Incognito

deon

Deon adds an ethics checklist to your data science projects.

About

Examples

- Data ethics in the real world

Rendered versions

- Markdown
- HTML
- Jupyter Notebook
- RST
- Text

 Check out deon on GitHub



Data ethics in the real world

To make the ideas contained in the checklist more concrete, we've compiled **examples** of times when tradoffs were handled well, and times when things have gone wrong. Examples are paired with the checklist questions to help illuminate where in the process ethics discussions may have helped provide a course correction. Positive examples show how principles of **deon** can be followed in the real world.

Checklist Question	Examples
Data Collection A.1 Informed consent: If there are human subjects, have they given informed consent, where subjects affirmatively opt-in and have a clear understanding of the data uses to which they consent?	 A voiceover studio is now required to get informed consent from a performer before using their likeness in AI-generated content.  Facebook uses phone numbers provided for two-factor authentication to target users with ads.  African-American men were enrolled in the Tuskegee Study on the progression of syphilis without being told the true purpose of the study or that treatment for syphilis was being withheld.  OpenAI's ChatGPT memorized and regurgitated entire poems without

deon/examples/ethics.md at · +

github.com/drivendataorg/deon/blob/main/examples/ethics.md

Incognito

PreviewCodeBlame

36 lines (29 loc) · 4.03 KB

RawCopyDownloadEditDropdownMenu

Data Science Ethics Checklist

ethics checklist deon

A. Data Collection

- ☐ **A.1 Informed consent:** If there are human subjects, have they given informed consent, where subjects affirmatively opt-in and have a clear understanding of the data uses to which they consent?
- ☐ **A.2 Collection bias:** Have we considered sources of bias that could be introduced during data collection and survey design and taken steps to mitigate those?
- ☐ **A.3 Limit PII exposure:** Have we considered ways to minimize exposure of personally identifiable information (PII) for example through anonymization or not collecting information that isn't relevant for analysis?
- ☐ **A.4 Downstream bias mitigation:** Have we considered ways to enable testing downstream results for biased outcomes (e.g., collecting data on protected group status like race or gender)?

B. Data Storage

- ☐ **B.1 Data security:** Do we have a plan to protect and secure data (e.g., encryption at rest and in transit, access controls on internal users and third parties, access logs, and up-to-date software)?
- ☐ **B.2 Right to be forgotten:** Do we have a mechanism through which an individual can request their personal information be removed?
- ☐ **B.3 Data retention plan:** Is there a schedule or plan to delete the data after it is no longer needed?

C. Analysis

- ☐ **C.1 Missing perspectives:** Have we sought to address blindspots in the analysis through engagement with relevant stakeholders (e.g., checking assumptions and discussing implications with affected communities and subject matter experts)?
- ☐ **C.2 Dataset bias:** Have we examined the data for possible sources of bias and taken steps to mitigate or address these biases (e.g.,

Key design perspectives



- **Our goal is not to be arbitrators of which ethical concerns merit inclusion.**
 - The default checklist is meant as a sensible starting point, and we believe teams will benefit from building custom checklists.

Key design perspectives



- Our goal is not to be arbitrators of which ethical concerns merit inclusion.
 - The default checklist is meant as a sensible starting point, and we believe teams will benefit from building custom checklists.
- **Checklist items are meant to provoke discussion.**
 - The goal of the checklist items are not to concretely recommend a specific action but rather are framed as prompts to discuss or consider.

Key design perspectives



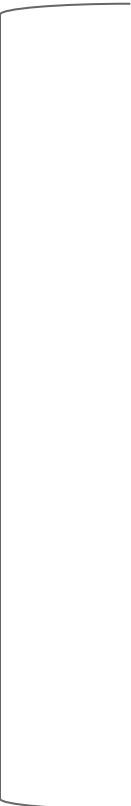
- Our goal is not to be arbitrators of which ethical concerns merit inclusion.
 - The default checklist is meant as a sensible starting point, and we believe teams will benefit from building custom checklists.
- Checklist items are meant to provoke discussion.
 - The goal of the checklist items are not to concretely recommend a specific action but rather are framed as prompts to discuss or consider.
- **Decisions on ethical courses of action are not up to data scientists alone.**
 - Checklist is designed to provoke conversations around issues where data scientists have particular responsibility and perspective to bring in stakeholders.

Key design perspectives



- Our goal is not to be arbitrators of which ethical concerns merit inclusion.
 - The default checklist is meant as a sensible starting point, and we believe teams will benefit from building custom checklists.
- Checklist items are meant to provoke discussion.
 - The goal of the checklist items are not to concretely recommend a specific action but rather are framed as prompts to discuss or consider.
- Decisions on ethical courses of action are not up to data scientists alone.
 - Checklist is designed to provoke conversations around issues where data scientists have particular responsibility and perspective.
- **Strictly technical best practices are not included.**
 - This is meant to be above and beyond statistical or technical correctness.

**Deon includes a
reasonable default
checklist.**



Data collection

Data storage

Analysis

Modeling

Deployment

Data collection

Informed consent

If there are human subjects, have they given informed consent, where subjects affirmatively opt-in and have a clear understanding of the data uses to which they consent?

Collection bias

Have we considered sources of bias that could be introduced during data collection and survey design and taken steps to mitigate those?

Limit PII exposure

Have we considered ways to minimize exposure of personally identifiable information (PII), for example through anonymization or not collecting information that isn't relevant for analysis?

Data collection

Informed consent

If there are human subjects, have they given informed consent, where subjects affirmatively opt-in and have a clear understanding of the data uses to which they consent?

Collection bias

Have we considered sources of bias that could be introduced during data collection and survey design and taken steps to mitigate those?

Limit PII exposure

Have we considered ways to minimize exposure of personally identifiable information (PII), for example through anonymization or not collecting information that isn't relevant for analysis?

Where things have gone wrong

StreetBump, a smartphone app to passively detect potholes, may fail to direct public resources to areas where smartphone penetration is lower, such as lower income areas or areas with a larger elderly population. [↗](#)

Data storage

Data security

Do we have a plan to protect and secure data (e.g., encryption at rest and in transit, access controls on internal users and third parties, access logs, and up-to-date software)?

Right to be forgotten

Do we have a mechanism through which an individual can request their personal information be removed?

Data retention plan

Is there a schedule or plan to delete the data after it is no longer needed?

Analysis

Missing perspectives

Have we sought to address blind spots in the analysis through engagement with relevant stakeholders (e.g., checking assumptions and discussing implications with affected communities and subject matter experts)?

Dataset bias

Have we examined the data for possible sources of bias and taken steps to mitigate or address these biases (e.g., stereotype perpetuation, confirmation bias, imbalanced classes, or omitted confounding variables)?

(cont.)

Where things have gone wrong

A 2024 study found that GPT-4 significantly and systematically over-exaggerated demographic variations in prevalence of medical conditions in the U.S. population. [↗](#)

Another 2025 study on asking an LLM for salary negotiation advice found that the salaries they recommended requesting were systematically lower for women and minorities. [↗](#)

Analysis

Honest representation

Are our visualizations, summary statistics, and reports designed to honestly represent the underlying data?

Privacy in analysis

Have we ensured that data with PII are not used or displayed unless necessary for the analysis?

Auditability

Is the process of generating the analysis well documented and reproducible if we discover issues in the future?

Analysis

Honest representation

Are our visualizations, summary statistics, and reports designed to honestly represent the underlying data?

Privacy in analysis

Have we ensured that data with PII are not used or displayed unless necessary for the analysis?

Auditability

Is the process of generating the analysis well documented and reproducible if we discover issues in the future?

Modeling

Proxy discrimination

Have we ensured that the model does not rely on variables or proxies for variables that are unfairly discriminatory?

Fairness across groups

Have we tested model results for fairness with respect to different affected groups (e.g., tested for disparate error rates)?

Metric selection

Have we considered the effects of optimizing for our defined metrics and considered additional metrics?

(cont.)

Modeling

Proxy discrimination

Have we ensured that the model does not rely on variables or proxies for variables that are unfairly discriminatory?

Fairness across groups

Have we tested model results for fairness with respect to different affected groups (e.g., tested for disparate error rates)?

Metric selection

Have we considered the effects of optimizing for our defined metrics and considered additional metrics?

(cont.)

Where things could have gone wrong

Amazon scrapped the development of an in-house automated resume screening tool that was found to penalize candidates who attended all-women's colleges or had terms like "women's chess club captain". This was a case where they successfully caught this problem before the tool was ever deployed. [↗](#)

Modeling

Proxy discrimination

Have we ensured that the model does not rely on variables or proxies for variables that are unfairly discriminatory?

Fairness across groups

Have we tested model results for fairness with respect to different affected groups (e.g., tested for disparate error rates)?

Metric selection

Have we considered the effects of optimizing for our defined metrics and considered additional metrics?

(cont.)

Where things have gone wrong

The COMPAS recidivism risk algorithm, the subject of the famous ProPublica investigation Machine Bias [↗](#), was tuned for equal accuracy between groups, but had a false positive rate for black people twice as high as for white people. This is a case where carefully considering a definition of fairness and choosing an appropriate metric is critical. [↗](#)

Modeling

Explainability

Can we explain in understandable terms a decision the model made in cases where a justification is needed?

Communicate bias

Have we communicated the shortcomings, limitations, and biases of the model to relevant stakeholders in ways that can be generally understood?

Where things have gone wrong

In a case study on hospital readmissions, patients with a history of asthma who were predicted by models to have a lower risk of death. It turns out this reflected the fact that they were usually admitted to intensive care units which led to better outcomes. A naive use of these predictions for these patients would have been unsafe. [↗](#)

Deployment

Redress

Have we discussed with our organization a plan for response if users are harmed by the results?

Roll back

Is there a way to turn off or roll back the model in production if necessary?

Concept drift

Do we test and monitor for concept drift to ensure the model remains fair over time?

Unintended use

Have we taken steps to identify and prevent unintended uses and abuse of the model and do we have a plan to monitor these once the model is deployed?

Where things have gone wrong

An AI agent set up as an experimental open source software contributor published a hit piece in retaliation against a volunteer maintainer who had rejected its contribution per the project's policy. [↗](#)

What did we talk about?

1. Why data ethics is (still) important today
2. How checklists and examples can help
3. Deon, an open source project for data ethics

Thank you!

Learn more at:
<https://deon.drivendata.org/>

DRIVENDATA